Задача №1

```
Load =4
Store 15
Sub =97
Store 4
Write =15
Write 15
Write *15
Halt 0
```

Задача №2

Вход: $a \geq 1$, $b \geq 1$. Выход: $a \bmod b$.

```
Read 1  ; r1 := a
Read 2  ; r2 := b
Load 1  ; r0 := a
Div 2   ; r0 := a // b
Mult 2  ; r0 := (a // b) * b
Sub 1   ; r0 := ((a // b) * b) - a
Mult =-1 ; r0 := (((a // b) * b) - a) * (-1) = a % b
Write 0  ; a % b -> OUT
```

Задача №3

Вход: $n \geq 1$, $a_1, \ldots, a_n$. Выход: $a_n, \ldots, a_1$.

Псевдокод

```
read n
for i = 0 to n - 1:
    A[i] := read()
for i = n - 1 downto 0:
    write(A[i])
```

Программа

```
Read 1  ; r1 := n
Load =3
Store 2  ; r2 := 3 = arr.begin()
Load 1
Add =3
Store 1  ; r1 := n + 3 = arr.end()
READ_LOOP: Load 1
 Sub 2
 JZero READ_DONE  ; if r2 == arr.end() then exit read
 Read *2
 Load 2
 Add =1
```

```
 Store 2    ; r2--
 Jump READ_LOOP
READ_DONE: Load =2
 Store 2    ; r2 := arr.begin() - 1
    Load 1
    Sub =1
    Store 1             ; r3 := &arr.last
OUTPUT_LOOP: Load 1
 Sub 2
 JZero END
 Write *1
 Load 1
 Sub =1
 Store 1    ; r1--
 Jump OUTPUT_LOOP
END: Halt 0
```